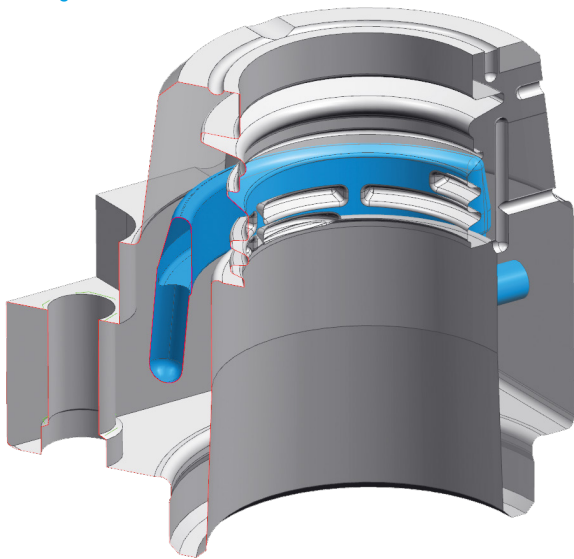


OH-XCOOL®

COOL DOWN.

OH-XCOOL® ACHIEVES TANGIBLE EFFICIENCY ENHANCEMENT IN THE PRODUCTION OF PET PREFORMS. THE IN-HOUSE DEVELOPMENTS OF OTTO HOFSTETTER AG SIGNIFICANTLY IMPROVE THE COOLING OF THREADS AND SUPPORT RINGS.

Diagram
of the OH-XCool®
cooling channel



It is common knowledge that in an injection mould, the first attempts strive to inject the material very quickly, without wasting any of the material, and to avoid any loss of temperature in the molten mass as much as possible. Once this process has

been completed, the next immediate, follow-on step in high-performance moulds like the ones needed, for instance, for the production of PET preforms is cooling of the injection-moulded part. The ever shorter cycle times are, therefore, dependent on a very even temperature in the mould and absolutely precision-adjusted process parameters in the control of the injection moulding machine. When a particularly high volume of plastic has to be cooled in the thread and support ring of the preform, and this is then the determining factor for cycle time, attention mainly focuses on this aspect.

Cooling in line with conventional standards.

What goes for many other types of moulds goes for the PET moulds manufactured by Otto Hofstetter AG, too. The mould components are fitted, if possible, with efficient cooling. The generally valid maxim in the construction of these steel parts is that the cooling channel should be positioned as close as possible to the mould contour of the injection-moulded part.

They are manufactured using standard operating procedures like, for instance, drilling or milling. In most cases, this is sufficient to achieve the required cooling capacity.

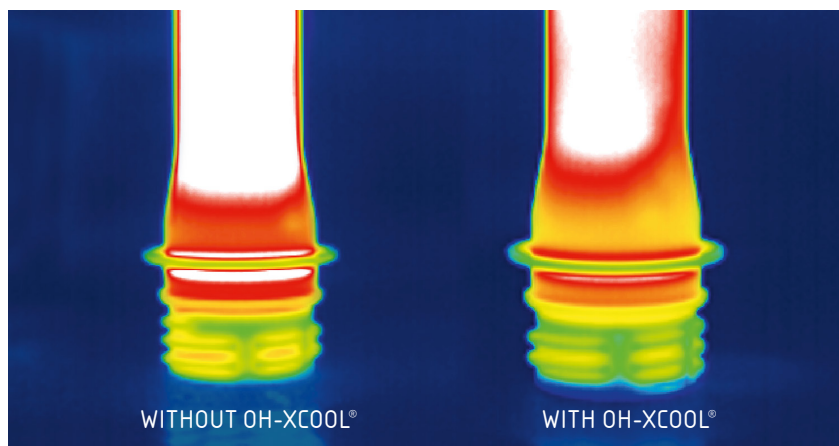
The thread takes things up a notch.

However, cooling in the thread insert of a PET mould imposes considerably stiffer and highly specific requirements for the production of cooling channels. The dimensions of these components are significantly smaller and the amount of steel to be processed as the basic material is available in a far more limited quantity. Consequently, only very small bore holes can be drilled. This leads to a limited cooling capacity and, in addition, to a far higher risk of contamination. Furthermore, optimum convergence with the mould contour is only partially possible.

OH-XCool® adds a new twist.

Basically, this is where the OH-XCool® technology developed by technicians in Uznach steps in. Thanks to a special manufacturing process, they have succeeded in integrating a wrap-around cooling channel close to the contour in an optimum position in terms of the thread and support ring. What is more, this exclusive method offers yet another advantage. Thanks to the position of the bore hole it is also possible to lay a wider channel, which is the precondition for a far higher cooling capacity. This leads to a reduction in cycle time which more than pays for itself. Depending on the type of preform, the process is shortened by up to 15 per cent. OH-XCool® is available as an option for all new generation moulds of Otto Hofstetter AG.

PREFORM / THREAD TYPE	WEIGHT	WALL THICKNESS	CYCLETIME	
			STANDARD	OH-XCOOL®
Bericap 30-37	34.0g	3.57 mm	14.6"	13.3"
Twist off	29.8g	2.70 mm	10.5"	10.0"
PCO 28 short	18.15g	2.47 mm	9.2"	8.3"
PCO 28 Alcoa	41.25g	3.71 mm	14.2"	12.5"



In the thermal images, a clear reduction in the "hot points" in the thread is visible.